We know that, the area of the parellelogram = 72 $sq.cm$

$$b \times h = 72$$

$$12 \times h = 72$$

$$h = \frac{72}{12} = 6\ cm$$

Therefore, the distance between the longer sides = 6 cm.

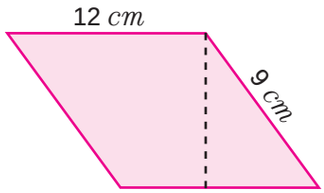


Fig.2.9

Example 2.5

The base of the parallelogram is thrice its height. If the area is 192 $sq.\ cm$, find the base and height.

Solution

Let the height of the parallelogram = $h\ cm$

Then the base of the parallelogram = $3h\ cm$

Area of the parallelogram = 192 $sq.\ cm$

$$b \times h = 192$$

$$3h \times h = 192$$

$$3h^2 = 192$$

$$h^2 = 64$$

$$h \times h = 8 \times 8$$

$$h = 8\ cm$$

$$\text{base} = 3h = 3 \times 8 = 24\ cm$$

Therefore, base of the parallelogram is 24 cm and height is 8 cm.

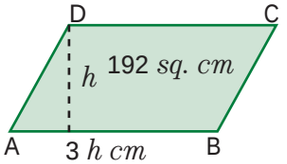


Fig.2.10

DO YOU KNOW?

Most of the bridges are constructed using parallelogram as the structural design (For example, Pamban Bridge in Rameswaram)

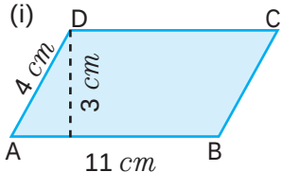
Engineers use properties of parallelograms to build and repair bridges.



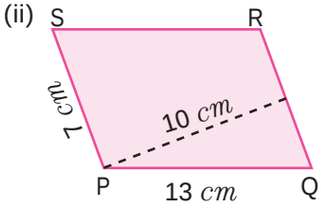
Exercise 2.1

1. Find the area and perimeter of the following parallelograms:

(i)



(ii)



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